Intermittent Adaptive Resilient Secure Control for Nonlinear CPSs Under DoS Attacks in Sensor Networks

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***Abstract*—This paper investigates the problem of adaptive prescribed-time intermittent resilient tracking control for a class of nonlinear cyber-physical systems (CPSs) under full-state denial-of-service (DoS) attacks with malicious intent. Since the disturbance of DoS attacks causes the discontinuity of the system states, which leads to its non-differentiability, it certainly increases the difficulty of the subsequent controller design. To overcome this obstacle, we reconstruct the system states by designing a set of adaptive high-gain observers, which generate differentiable state estimation signals. Subsequently, a time-varying constraint function is then integrated into the backstepping framework to construct a novel adaptive resilient controller with prescribed-time performance. This approach effectively converts the prescribed-time tracking control problem into a constraint problem for the tracking error. Meanwhile, a dynamic-threshold event-triggered mechanism (ETM) is proposed to minimize redundant signal transmissions and enhance the efficiency of network resource usage. Moreover, by using the Lyapunov stability theory, the proposed security control strategy ensures asymptotic convergence of the tracking error to zero, even in the absence of prior information regarding the frequency and duration of DoS attacks. Finally, the validity and practicality of the developed scheme are verified through simulation results on a one-link manipulator system.**

***Index Terms*—Nonlinear cyber-physical systems, event-triggered control, prescribed-time tracking control, denial-of-service attacks.**

I. Introduction

DURING the problems of tracking control for nonlinear systems have attracted much attention as a typical class of research topics, and many important results have been achieved [1], [2], [3], [4], [5]. With the widespread acceptance of the prescribed-time control, the tracking performance of the system has been significantly enhanced [6]. However, as time t tends to the predefined time T, this may appear as the infinite gain phenomenon. To further address this problem, a performance function-based partial tracking errors constrained control framework for nonlinear systems was constructed in [7]. In [8], an adaptive prescribed-time dynamic surface control method was proposed for a differential-drive autonomous surface vehicle with completely unknown parameters. In addition, the challenge of tracking the optimal blade tip speed ratio for maximum power within the defined time was conquered in [9], which in turn increased the annual power production of the variable-speed wind turbines.

Over the recent years, the network communication technology has played a weighty role in industrial production and its economic value is incalculable. Nevertheless, the information transmission in the networks is often subjected to intentional damage from the enemies, which is called cyber-attacks [10]. In order to protect the security of social property, substantial research efforts have been devoted to cyber-attack defense, with security control strategies for denial-of-service (DoS) attacks emerging as a prominent area of study [11]. In practice, DoS attacks mostly occur in the intermittent manner due to the limited resources of the adversaries [12]. According to this situation, a natural attack model was constructed [13] as well as used in [14] and [15] to obtain some results. Furthermore, when DoS attacks are active, the controller cannot obtain state variables continuously, leading to the inability to apply the traditional backstepping design methods. To cope with this challenge, the adaptive observer was designed [16]. The attack compensators were applied in [17] and [18], and the estimators were designed using the compensated signals. Existing results in [16], [17], and [18] assume prior knowledge of the attacker's duration and frequency data for launching an attack. However, in real scenarios, these attacker resource characteristics are usually not available. In view of this, it is of great practical importance to investigate a control method that does not require prior knowledge of DoS attacks' frequency and duration.

It is worth noting that the controllers designed in the above results are updated continuously, which may lead to a high frequency of controller updates, thereby consuming a significant amount of communication resources. However, in

Received 7 October 2025; revised 6 January 2026; accepted 26 January 2026. Date of publication 30 January 2026; date of current version 10 February 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62303287, Grant 62376170, and Grant U25A20455; in part by the Natural Science Foundation of Shandong Province under Grant ZR2024QF017; in part by the Youth Innovation Technology Project of Higher School in Shandong Province under Grant 2025KJH087; and in part by the Natural Science Foundation of Liaoning Province of China under Grant 2025-MS-013. The editor coordinating the review of this article was G. Pan. *(Corresponding author: Xinjun Wang.)*

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Digital Object Identifier 10.1109/TGCN.2026.3659499

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real-world scenarios, the resources allocated to the system are limited. Therefore, the research question of how to design a controller to reduce the consumption of network resources has practical applications. Fortunately, an event-triggered control (ETC) scheme was introduced in [19] to address this issue. The literature [20] was based on this control scheme, it was upgraded, and a new switching threshold ETC algorithm was proposed. A new adaptive ETC strategy was established in [21] by introducing higher-order Lyapunov functions as well as auxiliary functions. It should be highlighted that the event-triggered mechanism (ETM) methods designed in the aforementioned studies are static, meaning the scheduling threshold parameter is fixed. To achieve more efficient data transmission while maintaining control performance, [22] proposed the dynamic ETC strategy. This technique dynamically adjusts the threshold parameter by introducing a positive internal variable. In addition, different dynamic ETC algorithms are designed for nonlinear switching systems and nonlinear strict-feedback systems in [23] and [24], respectively. Nevertheless, the results in the above literature did not consider the impact of cyber-attacks and failed to achieve asymptotic tracking control. Consequently, it is urgent to propose a dynamic event-triggered adaptive asymptotic tracking control algorithm for the nonlinear systems under cyber-attacks.

Through the discussion of the above results, we concentrate on the problem of asymptotic tracking performance guaranteed by intermittent adaptive resilient control for a class of nonlinear cyber-physical systems (CPSs) under DoS attacks. All states in the system are potentially vulnerable to DoS attacks during transmission. The main strengths of this article are shown below:

1) A set of adaptive high-gain filters are built to reconstruct the whole system states and generate smooth differentiable estimates so as to overcome the non-differentiability of state variables caused by DoS attacks. Furthermore, in contrast to [16], [17], and [18], our adaptive high-gain approach does not rely on a priori knowledge of DoS attacks' frequency and duration during controller design. It can be readily concluded that the frequency and duration of DoS attacks can be arbitrary (as long as they remain bounded), which significantly enhances the robustness of the system against malicious DoS attacks.
2) In this study, an adaptive control strategy with guaranteed prescribed-time performance is proposed for a class of uncertain nonlinear systems by embedding a dynamic constraint function. The convergence time and tracking precision can be explicitly preassigned by the user, independent of initial system states and parameter settings. Distinct from the control approaches in [20], [21], and [25], which only achieve ultimate boundedness, the method developed herein ensures that the tracking error diminishes to zero asymptotically as time progresses.
3) Compared with the ETC strategy proposed in [26], the ETC mechanism proposed in this paper not only saves communication resources, but also eliminates the need for the controller to access the remaining items of the ETM. It is worth noting that this approach provides a flexible framework for addressing the event-triggered adaptive security control problem.

II. Mathematical Preliminaries and Problem Formulation

A. Necessary System Descriptions

The nonlinear CPS considered in this article is modeled by the following differential equation:

$$\begin{cases} \dot{x}_m = x_{m+1} + \epsilon_m(\bar{x}_m), \\ \dot{x}_n = u + \epsilon_n(\bar{x}_n), \\ y = x_1, \end{cases} \tag{1}$$

where $m = 1, \ldots, n-1$. Let $\bar{x}_m$ be the state vector of system with $\bar{x}_m = [x_1, \ldots, x_m]^T \in \mathsf{R}^m$. $y \in \mathsf{R}$ is the output of the system and $u \in \mathsf{R}$ is the designed control input. Furthermore, $\epsilon_m(\star)$ is known smooth nonlinear function.

Regarding system (1), in the absence of a priori knowledge about the characteristics of DoS attacks, the goal of this paper is to construct a security controller that output $y(t)$ tracking to the desired trajectory $y_r(t)$ within a user-specified time T, while also guaranteeing that $\lim_{t\to+\infty} \|y(t) - y_r(t)\| = 0$ holds. Additionally, all signals in the closed-loop system are proven to remain bounded, and Zeno behavior is explicitly ruled out.

Lemma 1 ([27]): On the interval $[0, \infty)$, if the function ϵ is continuous and the inequality $\lim_{t\to+\infty} \int_0^t \epsilon(t)dt < \vec{\epsilon} < \infty$ holds with $\vec{\epsilon}$ being a constant, which in turn makes $\lim_{t\to+\infty} \epsilon(t) = 0$ holds.

Assumption 1: The reference signal y_r is n-order differentiable.

Assumption 2: The function ϵ_m satisfies the global Lipschitz condition, for any $\mathcal{X}_1, \mathcal{X}_2 \in \mathsf{R}^m$ such that

$$|\epsilon_m(\mathcal{X}_1) - \epsilon_m(\mathcal{X}_2)| \le \mathcal{L}_m \|\mathcal{X}_1 - \mathcal{X}_2\|,$$

where $\mathcal{L}_m$ is a known Lipschitz constant.

Remark 1: Assumption 1 is necessary due to the fact that repeat differentiation of the virtual controls occurs during the subsequent backstepping of the design [28], [29]. In addition, **Assumption 2** is a common standard assumption in observer design of nonlinear systems [30], [31].

B. Intermittent DoS Attacks

As network defense techniques continue to advance, attackers are gradually shifting from traditional DoS attacks to more stealthy and threatening intermittent attack strategies. Inspired by this attack pattern, we consider the impact of intermittent DoS attacks on the security for a class of nonlinear systems in this paper. The attackers achieve the goal of destabilizing the system by discretely and maliciously jamming each communication channel from the sensor to the controller.

For more convenient description of DoS attacks in the paper, let $\mathfrak{A}(\pi, t)$ denote the set of time instants in the time interval $[\pi, t]$, where at least one channel is attacked. Based on this, we let $\mathfrak{S}(\pi, t)$ represent the set of time instants in the time interval $[\pi, t]$ where all channels are secure. Immediately after that, define $|\mathfrak{A}(\pi, t)|$ and $|\mathfrak{S}(\pi, t)|$ to denote the total durations

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of being attacked and secure on $[\pi, t]$, respectively. In addition, $\{t_j^m\}_{j\in\mathcal{N}}$ represents the OFF/NO instants of DoS attack on the mth channel, where $m = 1, \ldots, n$. Define $t_j \subset t_j^m (j \in \mathcal{N})$ to denote the switching moment when the system is in attack-free or under-attack state.

Assumption 3 (DoS Duration [17]): There exists the constants $\mathcal{T} > 1$ and $\Bbbk > 0$ such that the following inequality holds:

$$|\mathfrak{A}(\pi, t)| \leq \frac{t-\pi}{\mathcal{T}} + \Bbbk, \tag{2}$$

where $0 \leq \pi \leq t$.

Assumption 4 (DoS Frequency [17]): There exist constants $\eta \geq 0$ and $\pi_D > 0$ satisfying

$$n(\pi, t) \leq \eta + \frac{t-\pi}{\pi_D}. \tag{3}$$

Remark 2: Unlike [16], [17], and [18], the aforementioned parameters $\Bbbk, \eta, \pi_D$ and $\mathcal{T}$ are only used in this paper for convergence analysis and are not involved in the construction of the controller. Moreover, when $\pi_D \to 0$ or $\mathcal{T} = 1$, it is easy to see that each transmission of the sensors may be hindered by the attacker, rendering the system perpetually open-loop. Therefore, the implementation of *Assumptions 3* and 4 to bound the maximum duration and frequency of DoS attacks is essential to keep the system (1) controllable. Similar bounded attack conditions are discussed in [13] and [32].

C. Dynamic Event-Triggered Mechanism

In order to reduce the communication burden and improve resource utilization, the following dynamic ETM is constructed in this paper:

$$\begin{aligned} &u(t) = \upsilon(t_k), \forall t \in [t_k, t_{k+1}), k \in \mathcal{N} \\ &t_{k+1} = \inf\{t > t_k | |e_u(t)| \geq \gamma_1 |\upsilon(t)| + \gamma_2 e^{-ct}\}, \end{aligned} \tag{4}$$

where $u(t)$ is the sampled control input, $\upsilon(t)$ is the continuous control signal being designed and $e_u = \upsilon - u$ is the event-triggered error of system (1). Moreover, γ_1, γ_2 and c are the designed positive constants, where γ_1 satisfies $0 \leq \gamma_1 < 1$.

III. Main Results

A. Adaptive High-Gain Filter Design

For the convenience of the subsequent backstepping design, we indicate whether the DoS attacks occur on the mth channel of the system (1) by constructing an auxiliary function $\rho_m(t)$, which satisfies

$$\rho_m(t) = \begin{cases} 1, & \text{the } m\text{th channel is not attacked,} \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Since the mth-subsystem ($m = 1, \ldots, n$) is attacked, in order to reconstruct the unmeasurable state variables, the β_mth-order adaptive higher-order filter is designed to be

$$\begin{cases} \dot{\hat{x}}_{m,1} = \hat{x}_{m+1,1} + \epsilon_m(\hat{\bar{x}}_m) - \ell_m(\hat{x}_{m,1} - \hat{x}_{m,2}), \\ \dot{\hat{x}}_{m,2} = \hat{x}_{m+1,1} + \epsilon_m(\hat{\bar{x}}_m) - \ell_m(\hat{x}_{m,2} - \hat{x}_{m,3}), \\ \qquad \cdots \\ \dot{\hat{x}}_{m,\beta_m} = \hat{x}_{m+1,1} + \epsilon_m(\hat{\bar{x}}_m) - \rho_m \ell_m(\hat{x}_{m,\beta_m} - x_m), \\ \hat{x}_m = \hat{x}_{m,1},\ \beta_m := n - m + 1, \end{cases} \tag{6}$$

where $\hat{x}_m$ is the estimate of x_m, $\hat{\bar{x}}_m = [\hat{x}_{1,1}, \ldots, \hat{x}_{m,1}]^T \in \mathsf{R}^m$, $\hat{x}_{n+1,1} = u$. Besides, ℓ_m is the adaptive higher-order gain and is designed as the following form:

$$\begin{cases} \dot{\ell}_{m,i} = \kappa_m(\ell_{m,i+1} - \ell_{m,i}), \\ \dot{\ell}_{m,\beta_m} = \rho_m \kappa_m \|\vec{\hat{x}}_m - \mathbf{1}_{\beta_m} x_m\|^2, \\ \ell_m = \ell_{m,1}, \end{cases} \tag{7}$$

where $i = 1, \ldots, \beta_m - 1$ and κ_m is the design parameter with $\kappa_m > 0$. $\vec{\hat{x}}_m = [\hat{x}_{m,1}, \ldots, \hat{x}_{m,\beta_m}]^T \in \mathsf{R}^{\beta_m}$. Furthermore, the initial condition $\ell_{m,i}(0) = 0, i = 1, \ldots, \beta_m$.

Let $e_m = \hat{x}_{m,i} - x_m$ be an estimate error and $\Delta\epsilon_m = \epsilon_m(\hat{\bar{x}}_m) - \epsilon_m(\bar{x}_m)$. By considering (1) and (6), the following equation can be obtained:

$$\begin{cases} \dot{e}_{m,1} = e_{m+1,1} + \Delta\epsilon_m - \ell_m(e_{m,1} - e_{m,2}), \\ \dot{e}_{m,2} = e_{m+1,1} + \Delta\epsilon_m - \ell_m(e_{m,2} - e_{m,3}), \\ \qquad \cdots \\ \dot{e}_{m,\beta_m} = e_{m+1,1} + \Delta\epsilon_m - \rho_m \ell_m e_{m,\beta_m}, \end{cases} \tag{8}$$

and define $e_m = [e_{m,1}, \ldots, e_{m,\beta_m}]^T$. Further, we can get

$$\dot{e}_m = \mathbf{1}_{\beta_m}(e_{m+1,1} + \Delta\epsilon_m) + \ell_m \Pi_m^\rho e_m, \tag{9}$$

where $\mathbf{1}_{\beta_m} = [1, \ldots, 1]^T \in \mathsf{R}^{\beta_m}$ and

$$\Pi_m^\rho = \begin{bmatrix} -1 & 1 & & \\ & \ddots & \ddots & \\ & & -1 & 1 \\ & & & -\rho_m \end{bmatrix}_{\beta_m \times \beta_m}. \tag{10}$$

Theorem 1: Consider the system (1) under DoS attacks and satisfying **Assumptions 1**–4. The adaptive high-gain filter described in (6) is capable of asymptotically approximating the system state variables, such that $\lim_{t\to+\infty} |\hat{x}_m(t) - x_m(t)| = 0$. In the meantime, the estimate $\hat{x}_m(t)$ possesses continuous derivatives up to the β_mth-order for all $t \geq 0$.

Proof: Define the following Lyapunov function:

$$\Omega_m^\rho = e_m^T P_m^\rho e_m, \tag{11}$$

where $\rho = 0$ or 1.

By giving ρ, we can further obtain

$$\begin{aligned} \dot{\Omega}_m^\rho &= 2e_m^T P_m^\rho [\mathbf{1}_{\beta_m}(e_{m+1,1} + \Delta\epsilon_m) + \ell_m \Pi_m^\rho e_m] \\ &\leq e_m^T [\ell_m P_m^\rho \Pi_m^\rho + \ell_m (P_m^\rho \Pi_m^\rho)^T + 2P_m^{\rho T} P_m^\rho] e_m \\ &\quad + \sum_{i=1}^{m} \mathcal{L}_i^2 \|e_i\|^2 + \left\| e_{m+1,1} \right\|^2. \end{aligned} \tag{12}$$

Then, we construct the total Lyapunov function $\Omega_\rho = \sum_{m=1}^n \Omega_m^\rho$ and combine the above formulae to derive the following inequality:

$$\dot{\Omega}_\rho \leq \sum_{m=1}^{n} e_m^T \Lambda_m^\rho e_m, \tag{13}$$

where $\Lambda_m^\rho = \ell_m P_m^\rho \Pi_m^\rho + \ell_m (P_m^\rho \Pi_m^\rho)^T + 2P_m^{\rho T} P_m^\rho + [(n - m + 1)\mathcal{L}_m^2 + 1]I$.

Here, we analyze (13) based on the fact that it operates in two different time intervals:

Case i) The system is maliciously attacked when $t \in \mathfrak{A}(0, +\infty)$, indicating that there is at least one index $m \in$

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$\{1,\ldots,n\}$ for which $\rho_m = 0$. This implies that there exist a positive definite matrix P_m^0 and a positive constant ω_0 such that the following inequalities hold:

$$P_m^0\Pi_m^0 + (P_m^0\Pi_m^0)^T \le 0,$$
$$\begin{pmatrix} -\omega_0 P_m^0 + [(n-m+1)\mathcal{L}_m^2 + 1]I & P_m^0 \\ P_m^0 & -\frac{1}{2}I \end{pmatrix} < 0. \tag{14}$$

Then, it follows from (13) that $\dot{\Omega}_\rho \le \omega_0\Omega_\rho$.

Case ii) When $t \in \mathfrak{S}(0,+\infty)$, $\rho_m = 1$ for all $m = 1,\ldots,n$. The matrix Π_m^1 is chosen as Huritz, it follows that for any given positive definite matrix Q_m, there exists a positive definite matrix P_m^1 such that $-Q_m = P_m^1\Pi_m^1 + (P_m^1\Pi_m^1)^T$.

Define

$$\omega_1 = \frac{\ell_m\lambda_{\min}(Q_m) - 2\|P_m^1\|^2 - (n-m+1)\mathcal{L}_m^2 - 1}{\|P_m^1\|}.$$

In turn, the equation (13) ensures that $\dot{\Omega}_\rho \le \omega_1\Omega_\rho$ holds.

Moreover, there exists a positive constant $\varkappa > 1$ such that $P_m^0 \le \varkappa P_m^1$ and $P_m^1 \le \varkappa P_m^0 (m = 1,\ldots,n)$. Consider the time $t \in [t_j, t_{j+1})$, based on the above analysis and by iterating over ***Cases i and ii***, it is obtained that

$$\begin{aligned}
\Omega_\rho(t) &\le e^{\omega_{t_j}(t-t_j)}\Omega_\rho(t_j) \\
&\le \varkappa e^{\omega_{t_j}(t-t_j)}\Omega_\rho(t_j^-) \\
&\le \varkappa e^{\omega_{t_j}(t-t_j)+\omega_{t_{j-1}}(t-t_{j-1})}\Omega_\rho(t_{j-1}) \\
&\le \varkappa^2 e^{\omega_{t_j}(t-t_j)+\omega_{t_{j-1}}(t-t_{j-1})}\Omega_\rho(t_{j-1}^-) \\
&\cdots \\
&\le \varkappa^{n(0,t)} e^{\omega_0\mathfrak{A}(0,t)-\omega_1\mathfrak{S}(0,t)}\Omega_\rho(0) \\
&\le \varkappa^{\eta} e^{(\omega_0+\omega_1)\mathbb{k}} e^{-ht}\Omega_\rho(0),
\end{aligned} \tag{15}$$

where $h = \frac{\mathcal{T}-1}{\mathcal{T}}\omega_1 - \frac{\log\varkappa}{\pi_D} - \frac{\omega_0}{\mathcal{T}}$.

For convenience in subsequent analyses, we define

$$\Theta_m = \frac{2\|P_m^1\|^2 + (n-m+1)\mathcal{L}_m^2 + 1 + \frac{\mathcal{T}\|P_m^1\|}{\mathcal{T}-1}\left(\frac{\log\varkappa}{\pi_D} + \frac{\omega_0}{\mathcal{T}}\right)}{\lambda_{\min}(Q_m)}.$$

With respect to the convergence of the error dynamics (9), we next analyse it in three separate cases:

Case 1: For any $t \in [0,+\infty)$, $\ell_{m,\beta_m} \le \Theta_m (m = 1,\ldots,n)$ is always satisfied. Based on ***Lemma 1***, we can easily obtain that $\lim_{t\to+\infty}\rho_m(t)\|\vec{x}_m(t) - \mathbf{1}_{\beta_m}x_m(t)\|^2 = 0$ holds. This implies that the error $e_m(t)$ converges to zero during the interval $\mathfrak{S}(0,+\infty)$.

Then, by analysing ***Case i)***, it is clear that for $\forall t \in \mathfrak{A}(0,+\infty)$, there is $\dot{\Omega}_\rho \le \omega_0\Omega_\rho$. We have

$$\Omega_\rho(t) \le e^{\omega_0(t-t_j)}\Omega_\rho(t_j) \le \varkappa e^{\omega_0(t-t_j)}\Omega_\rho(t_j^-),$$

where $j = \max_j\{t_j \le t\}$ and $t_j^- \in \mathfrak{S}(0,+\infty)$. By ***Assumption 4***, it is possible to obtain $t - t_j \le \frac{\mathbb{k}\mathcal{T}}{\mathcal{T}-1}$. In view of the above, it follows that

$$\Omega_\rho(t) \le e^{\omega_0(t-t_j)}\Omega_\rho(t_j) \le \varkappa e^{\frac{\omega_0\mathbb{k}\mathcal{T}}{\mathcal{T}-1}}\Omega_\rho(t_j^-) \tag{16}$$

Thus, by letting $t \to +\infty$, i.e., $t_j^- \to +\infty$, we have $\Omega_\rho(t) \to 0$. Eventually, the error $e_m(t)$ converges to zero during the interval $\mathfrak{A}(0,+\infty)$.

Case 2: For all $m = 1,\ldots,n$, there exists a time $\hat{t}_1$ satisfying $\ell_{m,\beta_m}(\hat{t}_1) > \Theta_m$. By analysing (7), we can know that $\ell_{m,\beta_m}(t)$ is monotonically increasing, which means that $\ell_{m,\beta_m}(t) > \Theta_m$ for all $t \ge \hat{t}_1$. Further, the existence of a small positive constant a_1 that makes $\ell_{m,\beta_m}(t) \ge \Theta_m + a_1$ holds true for all $t \ge \hat{t}_1$.

Based on the expression for $\dot{\ell}_{m,\beta_m-1}$, one gets

$$\begin{aligned}
\ell_{m,\beta_m-1}(t) &= \kappa_m\int_{\pi=0}^{t} e^{-\kappa_m(t-\pi)}\ell_{m,\beta_m}(\pi)d\pi \\
&\ge \kappa_m\int_{\pi=\hat{t}_1}^{t} e^{-\kappa_m(t-\pi)}\ell_{m,\beta_m}(\pi)d\pi \\
&\ge (\Theta_m + a_1)\kappa_m\int_{\pi=\hat{t}_1}^{t} e^{-\kappa_m(t-\pi)}d\pi \\
&\ge (\Theta_m + a_1)(1 - e^{-\kappa_m(t-\hat{t}_1)}).
\end{aligned} \tag{17}$$

With the help of the above analysis, it can be obtained that there exists a time $\hat{t}_2 > \hat{t}_1 + \frac{1}{\kappa_m}\log\frac{\Theta_m+a_1}{a_1}$ such that $\ell_{m,\beta_{m-1}}(t) > \Theta_m$ for any $t \ge \hat{t}_2$. Furthermore, by applying the above analysis to each subsystem $\dot{\ell}_{m,i}(i = \beta_m - 2,\ldots,1)$, we are able to access the existence of a time $\hat{t}'$ such that $\ell_m(t) > \Theta_m$ for $\forall t \ge \hat{t}'$. Based on the definition of Θ_m, one has $h > 0$ when $t \ge \hat{t}'$. From (15), as $t \to +\infty$, $\Omega_\rho(t) \to 0$ holds.

Case 3: This case can be analysed based on ***Case 1*** and ***Case 2***. For any $t \in [0,+\infty)$, it follows that $\ell_{m,\beta_m}(t) \le \Theta_m$ with $m \in \mathfrak{T} \subsetneq \{1,\ldots,n\}$. And for $m \in \mathfrak{F} = \{1,\ldots,n\}\backslash\mathfrak{T}$, there exists a time $\bar{t}$ such that $\ell_{m,\beta_m}(\bar{t}) > \Theta_m$. Let $\Omega_{\rho,\mathfrak{F}} = \sum_{m\in\mathfrak{F}}\Omega_m^\rho$, we have

$$\begin{aligned}
\dot{\Omega}_{\rho,\mathfrak{F}} &\le \sum_{m\in\mathfrak{F}} e_m^T\Lambda_m^\rho e_m + \sum_{m\in\mathfrak{T}}\|e_m\|^2 \\
&\le \omega_\rho\dot{\Omega}_{\rho,\mathfrak{F}} + \sum_{m\in\mathfrak{T}}\|e_m\|^2.
\end{aligned} \tag{18}$$

For the given time $t \in [t_j, t_{j+1})$, similar to equation (15), we have

$$\begin{aligned}
\Omega_{\rho,\mathfrak{F}}(t) &\le e^{\omega_{t_j}(t-t_j)}\Omega_{\rho,\mathfrak{F}}(t_j) + \int_{t_j}^{t} e^{\omega_{t_j}(t-\pi)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\le \varkappa e^{\omega_{t_j}(t-t_j)}\Omega_{\rho,\mathfrak{F}}(t_j^-) + \int_{t_j}^{t} e^{\omega_{t_j}(t-\pi)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\le \varkappa e^{\omega_{t_j}(t-t_j)+\omega_{t_{j-1}}(t_j-t_{j-1})}\Omega_{\rho,\mathfrak{F}}(t_{j-1}) \\
&\quad + \varkappa\int_{t_{j-1}}^{t_j} e^{\omega_{t_j}(t-t_j)+\omega_{t_{j-1}}(t_j-\pi)+}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\quad + \int_{t_j}^{t} e^{\omega_{t_j}(t-\pi)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\le \varkappa^{n(0,t)} e^{\omega_0\mathfrak{A}(0,t)-\omega_1\mathfrak{S}(0,t)}\Omega_{\rho,\mathfrak{F}}(0) \\
&\quad + \int_{\pi=0}^{t}\varkappa^{n(\pi,t)} e^{\omega_0\mathfrak{A}(\pi,t)-\omega_1\mathfrak{S}(\pi,t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\le \varkappa^{\eta} e^{(\omega_0+\omega_1)\mathbb{k}}\left[e^{-ht}\Omega_{\rho,\mathfrak{F}}(0) + \int_{\pi=0}^{t} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi\right].
\end{aligned} \tag{19}$$

It follows from ***Case 2*** that there exists a time $\hat{t}''$ such that $h > 0$ is satisfied. Immediately afterwards, we proved that $\lim_{t\to+\infty}\int_{\pi=0}^{t} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi = 0$ in the time interval $\mathfrak{S}(0,+\infty)$.

Furthermore, based on ***Case 1***, it can be deduced that $\sum_{m\in\mathfrak{T}}\|e_m(t)\|^2$ converges to zero in the time interval $\mathfrak{S}(0,+\infty)$. Subsequently, for arbitrary given $\varrho > 0$, one can identify a finite time $\mathfrak{T}_\varrho$ such that the inequality

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$\sum_{m\in\mathfrak{T}}\|e_m(t)\|^2 < \frac{\varrho h}{2}$ holds for all $t \in \mathfrak{S}(\mathfrak{T}_\varrho, \infty)$. In addition, there is a positive constant $\wp$ such that $\sum_{m\in\mathfrak{T}}\|e_m(t)\|^2 \le \wp$ for $t \in \mathfrak{S}(0, +\infty)$. Therefore, within the interval $t \in \mathfrak{S}(\mathfrak{T}_\varrho, +\infty)$, we can get

$$\begin{aligned}
&\int_{\mathfrak{S}(0,t)} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&= \int_{\mathfrak{S}(0,\mathfrak{T}_\varrho)} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\quad + \int_{\mathfrak{S}(\mathfrak{T}_\varrho,t)} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi \\
&\le \frac{\wp}{h}e^{-h(t-\mathfrak{T}_\varrho)} + \frac{\varrho}{2}.
\end{aligned} \tag{20}$$

When $t > \max\left\{\hat{t}', \mathfrak{T}_\varrho + \frac{1}{h}\ln\frac{2\wp}{h\varrho}\right\}$, the following inequality holds:

$$\int_{\pi=0}^{t} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi < \varrho. \tag{21}$$

As a result of the above analysis, it is proven that $\int_{\pi=0}^{t} e^{h(\pi-t)}\sum_{m\in\mathfrak{T}}\|e_m(\pi)\|^2 d\pi$ converges to zero during the time interval $\mathfrak{S}(0, +\infty)$. As well, $\Omega_{\rho,\mathfrak{F}}(t)$ and $\Omega_\rho(t)$ converge to zero in the interval $\mathfrak{S}(0, +\infty)$. The analysis of the convergence on the time interval $\mathfrak{A}(0, +\infty)$ is similar to that of ***Case 1***, which ultimately yields that $\Omega_\rho(t)$ also converges to zero on the time interval $\mathfrak{A}(0, +\infty)$. Moreover, based on the structural feature of the filter (6), it is not difficult to obtain the first order derivative of the existence estimator $\hat{x}_m$. ∎

Remark 3: In contrast to the literatures [16], [17], [18], the proposed adaptive higher-order filter in equation (6) achieves the accurate estimations of the system state without needing a priori information about the attack-related parameters π_D and $\mathcal{T}$. Furthermore, to further ensure that the state estimates possess the necessary higher-order differentiability, the signal $\rho_m(t)$ is generated using the recursive high-order filtering procedure detailed in (7).

B. Adaptive Prescribed-Time Security Controller Design

The tracking error of the nonlinear system (1) is defined as

$$z_1 = \hat{y} - y_r, \tag{22}$$

where $\hat{y} = \hat{x}_1$.

Given the user-specified convergence time interval $T \in (0, +\infty)$ and desired tracking accuracy $\sigma \in (0, +\infty)$, the following prescribed-time constraint function can be built:

$$\mathcal{G}(t) = \begin{cases} \sigma, & t > T, \\ \left(\frac{1}{t} - \frac{1}{T}\right)^{2\jmath} + \sigma, & 0 \le t \le T, \end{cases} \tag{23}$$

where $\jmath$ is a positive integer. In order to guarantee the continuous differentiability of $\mathcal{G}^{(a)}(t)$, $a = 0, 1, \ldots, n$, where n denotes the system's order, and we set $2\jmath > n + 1$.

For ensuring that the tracking control objective is achieved within the prescribed-time, the tracking error must satisfy the constraints imposed below:

$$-\mathcal{G}(t) \le z_1(t) \le \mathcal{G}(t). \tag{24}$$

By the characterisation of $\mathcal{G}(t)$ involved in (23), we can get from (24) that

$$|z_1(t)| < \sigma, \quad t \ge T, \tag{25}$$

where z_1 is guaranteed to enter the target interval σ within the predefined time T by satisfying (25).

Since the constraint boundary in (23) is unbounded with respect to the initial time of the system, we proceed to propose the following constraint handling control strategy:

$$\xi(\nu(t)) = \begin{cases} -\left(\frac{\nu(t)}{\iota} - 1\right)^{2\jmath} + 1, & 0 < \nu(t) \le \iota, \\ 1, & \nu(t) > \iota, \end{cases} \tag{26}$$

where $\nu(t) = \mathfrak{b}(\mathcal{G}(t) - z_1(t))(\mathcal{G}(t) + z_1(t)) = \mathfrak{b}[\mathcal{G}^2(t) - z_1^2(t)]$. ι and $\mathfrak{b}$ represent the positive parameters designed by the user. It is worth noting that $\xi(\cdot)$ in (26) is a smooth composite function. Moreover, when $\nu(t) = \iota$, ξ is continuous. At the same time, the function $\xi(\cdot)$ is used to transform $\nu(t)$ into the interval $(0, 1]$.

Accordingly, the following coordinate transformation is introduced in the controller design:

$$\mathcal{H}_1(t) = \frac{z_1(t)}{\xi(\nu(t))} = \frac{z_1(t)}{\xi(\mathfrak{b}(\mathcal{G}^2(t) - z_1^2(t)))}. \tag{27}$$

Lemma 2 ([33]): If $\mathcal{H}_1 \in \mathbb{L}_\infty$, thereby $-\mathcal{G}(t) < z_1 < \mathcal{G}(t)$ holds for $\forall t > 0$, else $z_1 \to \mathcal{G}(t)$ or $z_1 \to -\mathcal{G}(t)$.

According to (27), the following coordinate transformation is formulated:

$$\zeta_1 = \mathcal{H}_1, \quad \zeta_m = \hat{x}_m - \alpha_{m-1}, \quad m = 2, \ldots, n. \tag{28}$$

Step 1: Based on the n adaptive high-order filters by (6) and (7) as well as the coordinate transformation (28), we have

$$\begin{aligned}
\dot{\zeta}_1 &= \aleph_1\dot{z}_1 + \chi_1 \\
&= \aleph_1(\hat{x}_2 - \ell_1(\hat{x}_{1,1} - \hat{x}_{1,2}) + \epsilon_1(\hat{\tilde{x}}_1) - \dot{y}_r) + \chi_1,
\end{aligned} \tag{29}$$

where

$$\aleph_1 = \begin{cases} \frac{1}{\xi} - \frac{4\mathfrak{b}\jmath}{\iota\xi^2}\left[\frac{\nu}{\iota} - 1\right]^{2\jmath-1} z_1^2 > 0, & 0 < \nu \le \iota, \\ 1, & \nu > \iota, \end{cases} \tag{30}$$

$$\chi_1 = \begin{cases} \frac{4\mathfrak{b}\jmath}{\iota\xi^2}\left[\frac{\nu}{\iota} - 1\right]^{2\jmath-1}\dot{\mathcal{G}}\mathcal{G}z_1, & 0 < \nu \le \iota, \\ 0, & \nu > \iota. \end{cases} \tag{31}$$

Select the Lyapunov function candidate as $\mathbb{V}_1 = \frac{1}{2}(\zeta_1)^2$. Then, differentiating $\mathbb{V}_1$ with respect to time shows

$$\begin{aligned}
\dot{\mathbb{V}}_1 &= \zeta_1\aleph_1(\hat{x}_2 - \ell_1(\hat{x}_{1,1} - \hat{x}_{1,2}) + \epsilon_1(\hat{\tilde{x}}_1) - \dot{y}_r) + \zeta_1\chi_1 \\
&= \zeta_1\aleph_1(-\ell_1(\hat{x}_{1,1} - \hat{x}_{1,2}) + \alpha_1 + \zeta_2 + \epsilon_1(\hat{\tilde{x}}_1) - \dot{y}_r) \\
&\quad + \zeta_1\chi_1.
\end{aligned} \tag{32}$$

Then, the virtual controller α_1 is constructed as

$$\alpha_1 = -\frac{b_1}{\aleph_1}(\zeta_1 + \chi_1) + \ell_1(\hat{x}_{1,1} - \hat{x}_{1,2}) - \epsilon_1(\hat{\tilde{x}}_1) + \dot{y}_r, \tag{33}$$

where $b_1 > 0$ is a design parameter.

Substituting (33) into (32) results in

$$\dot{\mathbb{V}}_1 = -b_1\zeta_1^2 + \aleph_1\zeta_1\zeta_2. \tag{34}$$

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Step j **($2 \le j \le n-1$):** Choose the Lyapunov function candidate as $\mathbb{V}_2 = \frac{1}{2}\zeta_2^2$. Its derivative can be further expressed as

$$\dot{\mathbb{V}}_2 = \zeta_2(\hat{x}_3 + \epsilon_2(\hat{\tilde{x}}_2) - \ell_2(\hat{x}_{2,1} - \hat{x}_{2,2}) - \dot{\alpha}_1), \tag{35}$$

where

$$\dot{\alpha}_1 = \sum_{i=1}^{2} \frac{\partial \alpha_1}{\partial \hat{x}_{1,i}}(\epsilon_1(\hat{\tilde{x}}_1) - \ell_1(\hat{x}_{1,i} - \hat{x}_{1,i+1}) + \hat{x}_2) + \frac{\partial \alpha_1}{\partial \ell_1}\kappa_1(\ell_{1,2} - \ell_{1,1}) + \sum_{i=0}^{1} \frac{\partial \alpha_1}{\partial y_r^{(i)}} y_r^{(i+1)}. \tag{36}$$

The virtual controller α_2 is chosen as

$$\alpha_2 = -b_2\zeta_2 + \ell_2(\hat{x}_{2,1} - \hat{x}_{2,2}) + \dot{\alpha}_1 - \epsilon_2(\hat{\tilde{x}}_2) - \aleph_1\zeta_1, \tag{37}$$

where $b_2 > 0$ is a know constant.

Substituting (37) into (35), one obtains that

$$\dot{\mathbb{V}}_2 = -b_2\zeta_2^2 - \aleph_1\zeta_1\zeta_2 + \zeta_2\zeta_3. \tag{38}$$

Step m **($3 \le m \le n-1$):** The Lyapunov function $\mathbb{V}_m$ is chosen as $\mathbb{V}_m = \frac{1}{2}\zeta_m$. Then, the time derivative of $\mathbb{V}_m$ becomes

$$\dot{\mathbb{V}}_m = \zeta_m(\hat{x}_{m+1} + \epsilon_m(\hat{\tilde{x}}_m) - \ell_m(\hat{x}_{m,1} - \hat{x}_{m,2}) - \dot{\alpha}_{m-1}), \tag{39}$$

where

$$\dot{\alpha}_{m-1} = \sum_{i=0}^{m-1} \frac{\partial \alpha_{m-1}}{\partial y_r^{(i)}} y_r^{(i+1)} + \sum_{j=1}^{m-1}\sum_{i=1}^{m-j} \frac{\partial \alpha_{m-1}}{\partial \ell_{j,i}}\kappa_j(\ell_{j,i+1} - \ell_{j,i}) + \sum_{j=1}^{m-1}\sum_{i=1}^{m-j+1} \frac{\partial \alpha_{m-1}}{\partial \hat{x}_{j,i}}(\epsilon_j(\hat{\tilde{x}}_j) - \ell_j(\hat{x}_{j,i} - \hat{x}_{j,i+1}) + \hat{x}_{j+1,1}). \tag{40}$$

The virtual controller α_{m-1} is designed as follows:

$$\alpha_{m-1} = -b_m\zeta_m + \ell_m(\hat{x}_{m,1} - \hat{x}_{m,2}) + \dot{\alpha}_{m-1} - \epsilon_m(\hat{\tilde{x}}_m) - \zeta_{m-1}, \tag{41}$$

where $b_m > 0$ is a know constant.

In conjunction with (39) and (41), we obtain

$$\dot{\mathbb{V}}_m = -b_m\zeta_m^2 - \zeta_m\zeta_{m-1} + \zeta_m\zeta_{m+1}. \tag{42}$$

***Step n*:** Choose the Lyapunov function $\mathbb{V}_n = \frac{1}{2}\zeta_n$. The derivative of $\mathbb{V}_n = \frac{1}{2}\zeta_n$ can be expressed as

$$\dot{\mathbb{V}}_n = \zeta_n\big(\upsilon - e_u + \epsilon_n(\hat{\tilde{x}}_n) - \rho_n\ell_n(\hat{x}_n - x_n) - \dot{\alpha}_{n-1} + \alpha_n - \alpha_n\big), \tag{43}$$

where

$$\dot{\alpha}_{n-1} = \sum_{j=1}^{n-1} \frac{\partial \alpha_{n-1}}{\partial \hat{x}_{j,n-j+1}}\big(\epsilon_j(\hat{\tilde{x}}_j) - \rho_j\ell_j(\hat{x}_{j,n-j+1} - x_j) + \hat{x}_{j+1,1}\big) + \sum_{j=1}^{n-1}\sum_{i=1}^{n-j} \frac{\partial \alpha_{n-1}}{\partial \hat{x}_{j,i}}\big(\epsilon_j(\hat{\tilde{x}}_j) - \ell_j(\hat{x}_{j,i} - \hat{x}_{j,i+1}) + \hat{x}_{j+1,1}\big) + \sum_{j=1}^{n-1}\sum_{i=1}^{n-j} \frac{\partial \alpha_{n-1}}{\partial \ell_{j,i}}\kappa_j(\ell_{j,i+1} - \ell_{j,i}) + \sum_{i=0}^{n-1} \frac{\partial \alpha_{n-1}}{\partial y_r^{(i)}} y_r^{(i+1)}. \tag{44}$$

The tuning function α_n is designed as follows:

$$\alpha_n = b_n'\zeta_n - \rho_n\ell_n(\hat{x}_n - x_n) + \epsilon_n(\hat{\tilde{x}}_n) - \dot{\alpha}_{n-1} + \zeta_n, \tag{45}$$

where b_n' is the positive number of the design.

Based on the above, we have

$$\dot{\mathbb{V}}_n = -b_n'\zeta_n^2 - \zeta_{n-1}\zeta_n + \zeta_n(\upsilon - e_u + \alpha_n). \tag{46}$$

The continuous control command is structured in the following form:

$$\upsilon = \frac{\zeta_n\alpha_n^2}{(1-\gamma_1)\sqrt{\zeta_n^2\alpha_n^2 + \delta^2}}, \tag{47}$$

where $\delta = e^{-dt}$ is a positive integral function with $d > 0$.

Based on the established dynamic ETM (4) and combined with (47), the following inequality can be derived:

$$-\zeta_n e_u \le \gamma_1|\zeta_n\upsilon| + |\zeta_n\gamma_2 e^{-ct}| \le -\gamma_1\zeta_n\upsilon + |\zeta_n||\gamma_2 e^{-ct}| \le -\gamma_1\zeta_n\upsilon + \frac{\zeta_n^2}{4} + \gamma_2^2 e^{-2ct}. \tag{48}$$

Substituting (47) and (48) into (46), we have

$$\dot{\mathbb{V}}_n \le -b_n\zeta_n^2 - \zeta_{n-1}\zeta_n + \gamma_2^2 e^{-2ct} + \delta, \tag{49}$$

where $b_n = b_n' + 1/4$.

Choose the global Lyapunov function $\mathbb{V} = \sum_{m=1}^{n}\mathbb{V}_m$. Combined with the above analysis, the following inequality holds:

$$\dot{\mathbb{V}} \le -\sum_{m=1}^{n} b_m\zeta_m^2 + \gamma_2^2 e^{-2ct} + \delta \le -b_0\mathbb{V} + \mathcal{B}, \tag{50}$$

where $b_0 = \min\{2b_m\}$ and $\mathcal{B} = \gamma_2^2 e^{-2ct} + \delta$.

C. System Stability

Theorem 2: Consider the nonlinear system (1) under DoS attacks that satisfies **Assumptions 1–4** as well as ***Lemmas 1-2***, the adaptive high-gain filter (6)-(7), the virtual controllers (33), (37), (45) and the continuous control command (47) can guarantee that the tracking error can reach zero as time approaches to infinity and all the signals of the closed-loop system are bounded. Additionally, the Zeno behavior caused by the dynamic ETM can be avoided.

Proof: Integrating (50) from 0 to t, then (50) can become the following one:

$$\mathbb{V}(t) \le -\sum_{m=1}^{n}\int_0^t b_m\zeta_m d\tau + \int_0^t \gamma_2^2 e^{-2c\tau}d\tau + \int_0^t \delta d\tau + \mathbb{V}(0). \tag{51}$$

By means of (51), we can derive the following inequality:

$$\sum_{m=1}^{n}\int_0^t b_m\zeta_m(\tau)d\tau \le \sum_{m=1}^{n}\left[\mathbb{V}(0) + \frac{1}{d} + \frac{\gamma_2}{2c}\right] < \infty. \tag{52}$$

Then, combining this with ***Lemma 1***, one can obtain $\lim_{t\to+\infty}\zeta_m = 0$. Further, by considering ***Theorem 1***, we can

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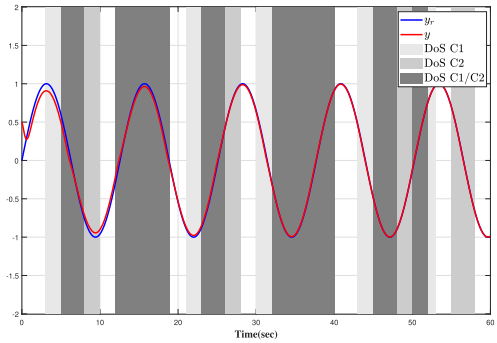


Fig. 1. Curve of out response y.

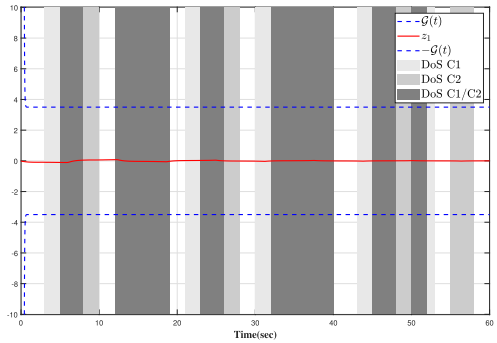


Fig. 2. Curve of tracking error z_1.

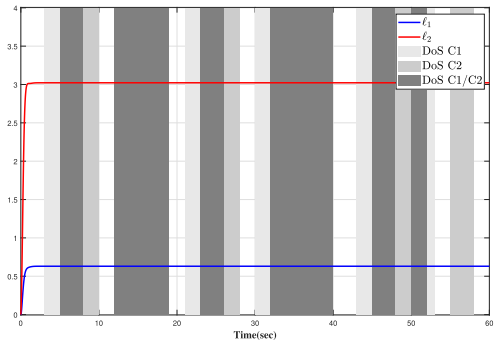


Fig. 3. Curves of adaptive parameters ℓ_1 and ℓ_2.

finally conclude that $\lim_{t\to+\infty}|y(t)-y_r(t)| = \lim_{t\to+\infty}|x_1(t)-\hat{x}_1(t)+\zeta_1| = 0$. Moreover, through the above proof process, we have $\mathbb{V} \in \mathbb{L}_\infty$. And because $\mathcal{H}_1$ is bounded, we can get $-\mathcal{G}(t) < z_1 < \mathcal{G}(t)$ by using Lemma 2. Then, by definition of $\mathcal{G}(t)$, it can be concluded that $|z_1| < \sigma$ for $t > T$.

Furthermore, by dynamic ETM (4) there is $e_u = \upsilon - u(t)$, $t \in [t_k, t_{k+1})$. In turn, we can obtain

$$\frac{d}{dt}|e_u| = \frac{d}{dt}\sqrt{e_u * e_u} = \text{sign}(e_u)\dot{e}_u \le |\dot{\upsilon}|. \tag{53}$$

Based on the above derivation process, we can deduce that $|\dot{\upsilon}|$ is bounded, due to the fact that $|\dot{\upsilon}|$ is a function of bounded signals. Hence, $|\dot{\upsilon}| \le W$ is satisfied, where W is a positive constant. Combining $e_u(t) = 0$ and $\frac{d}{dt}|e_u| \le W$, $|e_u| \le W(t-t_k)$ holds with $t \in [t_k, t_{k+1})$. Then, we can get

$$|e_u(t_{k+1})| = \gamma_1|\upsilon(t_{k+1})| + \gamma_2 e^{-ct_{k+1}}, \quad t = t_{k+1}. \tag{54}$$

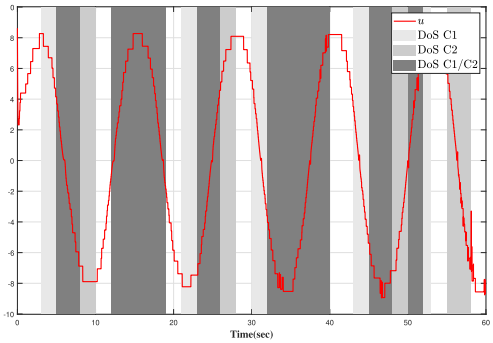


Fig. 4. Curve of control input u.

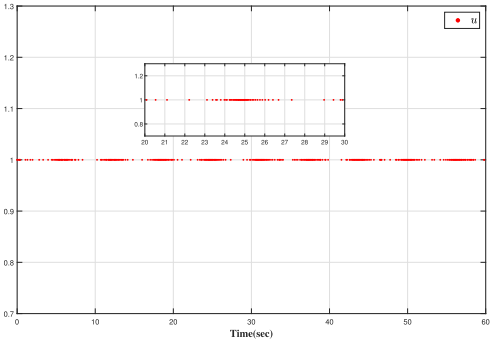


Fig. 5. Triggering times of the control input.

Thereby, one has

$$\gamma_2 e^{-ct_{k+1}} \le |e_u(t_{k+1})| \le W\bar{t}_k, \tag{55}$$

where $\bar{t}_k = t_{k+1} - t_k$.

It is straightforward to verify that $\bar{t}_k > 0$ holds for any finite interval. Here, we complete the proof by contradiction. Suppose $t_\infty = \lim_{k\to\infty} t_k < \infty$, which implies $\lim_{k\to\infty}\bar{t}_k = 0$. Using (55), it is possible to obtain $k \to \infty, 0 < \gamma_2 e^{-ct_\infty} \le W0 = 0$. Consequently, Zeno behavior is excluded. ■

IV. Simulation Studies

In this section, we consider the following single-link manipulator under DoS attacks:

$$\mathfrak{B}\ddot{o} + \mathfrak{Q}\dot{o} + \frac{1}{2}\mathfrak{M}gL\sin(o) = u(t), \tag{56}$$

where $x_1 = o$ is the angle of link, $x_2 = \dot{o}$ is the angular velocity, $\mathfrak{B}$ is the total rotational inertia of the motor, $\mathfrak{Q}$ is the total damping coefficient, L is the distance from the joint axis to the centre of mass, g is the acceleration of gravity and $\mathfrak{M}$ is the total mass of the link.

In the simulation, let $\mathfrak{B} = 1\text{kgm}^2$, $\mathfrak{Q} = 2\text{kgm}/s$, $g = 9.8\text{m}/s^2$, $L = 1\text{m}$, $\mathfrak{M} = 1\text{kg}$, the desired tracking accuracy $\sigma = 3.5$ and the predefined time $T = 1s$. The initial states are set as $x_1(0) = 0.1$ $x_2(0) = 0.2$. Then, we choose $\kappa_1 = 6$, $\kappa_2 = 8$ and $b_1 = 1$, $b_2 = 15$. The DoS attacks are randomly initiated with a total duration of $|\mathfrak{A}(0,60)| = 42s$ and occur $n(0,60) = 6$ times within the interval $[0, 60]$. Importantly, these parameters are not utilized in the controller design process. Additionally, the desired reference trajectory is specified as $y_r = \sin(0.5t)$

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The simulation results for the single-link robotic arm under DoS attacks are shown in Figs. 1–5. The grey part of the graph indicates that DoS attacks are active and the system (56) is under insecure environment. Fig. 1 depicts the trajectory of output y. The result presents in Fig. 2 illustrates that $|z_1(t)| < \sigma$ always holds at $t \geq T$. The trajectories of adaptive parameters ℓ_1, ℓ_2 and control input signal u are shown in Figs. 3–4. Moreover, Fig. 5 represents the triggering time of the system input, indicating that the execution interval is strictly positive.

V. Conclusion

This paper has proposed a dynamic event-triggered prescribed-time adaptive resilient tracking control strategy for a class of nonlinear CPSs subject to malicious DoS attacks. We utilized a set of adaptive high-gain filters established to produce the necessary state estimations to conquer the effects brought about by DoS attacks. In addition, by introducing a dynamic ETM, which in turn reduced the frequency of controller updates, the network communication resources were saved. On the basis of the above, a novel prescribed-time safety controller was presented by integrating a dynamic time-varying constraint function. The above control scheme not only guaranteed that the tracking was achieved in the predefined time but also reached the asymptotic convergence. It should be noted that the proposed control framework in this paper permits DoS attacks pattern with arbitrarily varying frequency and duration, so long as they remain bounded. Future research will aim to generalize this approach to nonlinear CPSs operating within more complex networked environments.

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